Blessed G/Michael Catholic Secondary School

Semester 1 Chemistry Final Examination for Grade11

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade11 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **2.** According to the Pauli Exclusion Principle, no two electrons in the same atom can have the same set of all four quantum numbers.

II. Choose the best answer among the given alternatives.

____ **1.** What is the wavelength (in meters) of the first line in the Lyman series for the hydrogen atom?

A. 36hc / (5RH) **B.** 4hc / (3RH) **C.** 56hc / (9RH) **D.** 4hc / RH

____ **3.** How many 3d electrons are present in the ground state electron configuration of a copper atom (Z = 29)?

A. 4 **B.** 5 **C.** 9 **D.** 10

____ **4.** How many unpaired electrons are there in the Cr3+ ion (Z = 24)?

A. 6 **B.** 3 **C.** 1 **D.** 0

____ **5.** Antimony (Sb) is a Group VA element. What are the molecular geometry and number of lone pairs on the central atom in the ion [SbCl4]-?

A. See-saw, 1 **B.** Square planar, 2

C. See-saw, 2 **D.** Linear, 3

____ **6.** What is the fundamental principle of Valence Bond Theory (VBT) for covalent bond formation?

A. Atomic orbitals merge to form delocalized molecular orbitals
B. Bonds form through the spatial overlap of half-filled atomic orbitals containing paired electrons
C. Electrons are delocalized evenly across the entire molecule
D. Hybridization is essential for all chemical bond formation

____ **7.** What is the hybridization and molecular geometry for the IF4- anion?

A. sp3d2, octahedral **B.** sp3d2, square planar

C. sp3, trigonal planar **D.** sp3d, trigonal bipyramidal

____ **8.** In Valence Bond Theory, which type of atomic orbital overlap results in the formation of a sigma (sigma) bond?

A. Side-by-side (lateral) overlap
B. No overlap is required for sigma bonds
C. Overlap involving only s-orbitals
D. End-to-end (head-on / axial) overlap along the internuclear axis

____ **9.** The hybridization of the central nitrogen atom in NO2+, NO3-, and NH4+ respectively are:

A. sp2, sp3 and sp **B.** sp2, sp and sp3

C. sp, sp2 and sp3 **D.** sp, sp3 and sp2

____ **10.** How many atomic orbitals have quantum numbers n = 5 and ml = +2?

A. 6 **B.** 1 **C.** 2 **D.** 3

____ **11.** The photon energies E1 and E2 of two electromagnetic radiations are 25 eV and 50 eV respectively. The relationship between their wavelengths lambda1 and lambda2 is:

A. lambda1 = lambda2 / 2 **B.** lambda1 = lambda2

C. lambda1 = 2 * lambda2 **D.** lambda1 = 4 * lambda2

____ **12.** The first five ionization energies of an element are 6.11, 11.87, 51.21, 67, and 84.37 eV respectively. The element is most likely:

A. 20Ca **B.** 19K **C.** 13Al **D.** 6C

____ **13.** Which intermolecular force is universally present in all molecules and atoms, including nonpolar species like liquid Argon (Ar)?

A. Hydrogen bonding **B.** Dipole-dipole interactions

C. London dispersion forces (LDF) **D.** Ion-dipole forces

____ **14.** Why are metallic elements characteristically malleable and ductile?

A. Metallic bonds are rigid and directional

B. The delocalized sea of valence electrons allows layers of metal cations to slide past one another without cleaving the bond

C. Metal atoms are very small and easily rearranged chemically

D. Metals have exceptionally high melting points

____ **15.** Which atomic factor generally increases the bonding strength of a metallic crystal lattice?

A. A larger atomic radius

B. Fewer delocalized valence electrons

C. A higher cationic charge and greater density of delocalized valence electrons

D. Weaker electrostatic attraction between ions and electrons

____ **16.** Which of the following statements is NOT true regarding the electron sea model of metallic bonding?

A. Electrons are static and individual valence electrons are localized to specific metal cations

B. Electrons are uniformly distributed throughout the metallic lattice

C. The nucleus of each atom is surrounded by outer electrons of all the atoms

D. Valence electrons are bound to the metal lattice by electrostatic attraction with cations

____ **17.** A single individual p-orbital can accommodate a maximum of how many electrons?

A. 6 electrons **B.** 3 electrons **C.** 2 electrons **D.** 1 electron

____ **18.** The three-dimensional region of space around the atomic nucleus where there is a high probability of finding an electron is called an:

A. Shell **B.** Orbital **C.** Energy level **D.** Bohr orbit

____ **19.** Which of the following molecules has a non-planar three-dimensional geometry?

A. CH4 **B.** BCl3 **C.** XeF4 **D.** HCHO

____ **20.** Which intermolecular phenomenon explains why at room temperature, I2 is a solid, Br2 is a liquid, and Cl2 is a gas?

A. Hydrogen bonding **B.** Ionic bonding

C. London dispersion forces **D.** Dipole-dipole forces

____ **21.** Which of the following physical characteristics is NOT a typical property of an ionic crystalline compound?

A. Malleable and ductile **B.** Brittle crystal structure

C. Hard physical integrity **D.** High melting and boiling points

____ **22.** What change in hybridization occurs at the sulfur atom in the reaction: SO2 + 1/2 O2 -> SO3?

A. sp2 -> sp3 **B.** sp3 -> sp3d

C. sp -> sp2 **D.** No hybridization change observed

____ **23.** Which of the following compounds is paired with an INCORRECT molecular geometry?

A. BF3, trigonal planar **B.** SF6, octahedral

C. NF3, trigonal planar **D.** CH4, tetrahedral

____ **24.** The maximum kinetic energy of photoelectrons emitted from a metallic surface is 1.03 * 10^-19 J when 656 nm light shines on it. What is the threshold frequency of the metal?

A. 4.57 * 10^14 s^-1 **B.** 3.02 * 10^14 s^-1

C. 3.02 * 10^-14 s^-1 **D.** 4.57 * 10^-14 s^-1

____ **25.** The threshold wavelength for photoelectric emission from a metal is 6500 Angstroms. The work function of the metal is approximately:

A. 2 eV **B.** 1 eV **C.** 0.1 eV **D.** 3 eV

____ **26.** Which pair of chemical species share the exact same molecular shape: NI3, I3-, SO3(2-), NO3-?

A. NI3 and I3- **B.** NI3 and SO3(2-)

C. SO3(2-) and NO3- **D.** I3- and NO3-

____ **27.** Consider the interhalogen cations and anions I. [ClF2]+ and II. [ClF2]-. Which geometric statement is correct?

A. I is bent, II is linear
B. I is linear, II is bent
C. Both are linear
D. Both are bent

____ **28.** Which molecule listed below possesses exactly two sigma (sigma) bonds and two pi (pi) bonds?

A. N2 **B.** C2H4 **C.** N2F2 **D.** HCN

____ **29.** Which of the following statements is NOT true regarding Bohr's postulates for the hydrogen atom?

A. The energy of an electron remains stationary as long as it occupies a permitted orbit
B. When electrons fall from an excited level to a lower level, they absorb a fixed quantum of energy
C. Electrons emit or absorb energy only when transitioning between discrete orbits
D. Electrons revolve around the nucleus in circular quantized orbits

____ **30.** Once formed, how do coordinate covalent (dative) bonds differ from ordinary covalent bonds?

A. They are noticeably stronger
B. They are more ionic in character
C. They are significantly weaker
D. There is no physical or chemical difference

____ **31.** Among CO2, H2O, BeCl2, and N2O, which species share the same linear molecular geometry?

A. CO2, H2O, and N2O
B. CO2 and BeCl2 only
C. CO2, BeCl2, and N2O
D. H2O and N2O only

____ **32.** Which chemical reaction involves a change in the electron pair geometry of the central atom?

A. NH3 + H+ -> NH4+
B. BF3 + F- -> BF4-
C. 2SO2 + O2 -> 2SO3
D. H2O + H+ -> H3O+

____ **33.** What is the de Broglie wavelength associated with an electron (m = 9.11 * 10^-31 kg) travelling at 40% of the speed of light?

A. 6.06 * 10^-10 m
B. 2.42 * 10^-10 m
C. 6.06 * 10^-12 m
D. 2.42 * 10^-12 m

____ **34.** Which of the following compounds exhibits strong intermolecular hydrogen bonding in the liquid state?

A. CH4 **B.** HCl **C.** H2S **D.** NH3

____ **35.** Which electronic transition emits the maximum photon energy in the hydrogen atom emission spectrum?

A. n = 4 -> n = 2 **B.** n = 4 -> n = 3 **C.** n = 2 -> n = 1 **D.** n = 3 -> n = 2

____ **36.** Which of the following is a characteristic physical property of nonpolar covalent compounds?

A. Extremely high melting and boiling points
B. Non-volatile crystalline salts
C. Rigid network ionic solids at room temperature
D. Generally low solubility in polar water

____ **37.** Formic acid (HCOOH) is secreted by ants. What hybridizations exist among the carbon and oxygen atoms in the molecule?

A. sp2 and sp3
B. sp and sp3
C. sp, sp2 and sp3
D. sp and sp2

____ **38.** Which atomic model first proposed that electrons revolve around the nucleus in discrete, stationary energy levels?

A. Plum Pudding Model
B. Bohr Model
C. Rutherford Model
D. Dalton Model

____ **39.** How many sigma (sigma) and pi (pi) bonds are present in 2-butene (H3C-CH=CH-CH3)?

A. 8 sigma and 1 pi
B. 10 sigma and 2 pi
C. 8 sigma and 2 pi
D. 11 sigma and 1 pi

____ **40.** Which of the following molecules does NOT possess a trigonal bipyramidal electron-pair geometry?

A. SF_4 **B.** ClF_3 **C.** XeF_2 **D.** BrF_5